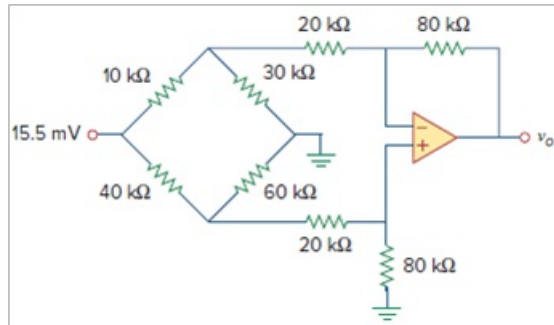


Problem

The circuit in Fig. 5.80 is a differential amplifier driven by a bridge. Find v_o .

Figure 5.80:



Step-by-step solution

Step 1 of 6

Refer to figure 5.80 in the textbook.

Redraw the circuit as shown in figure 1.

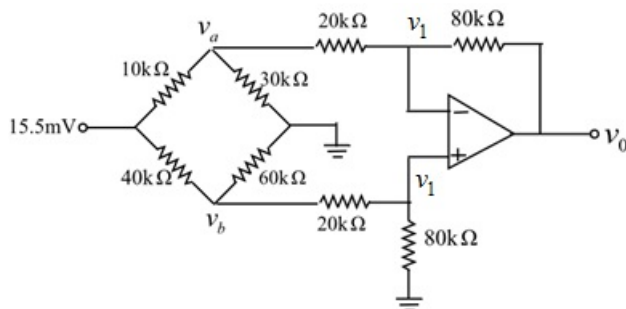


Figure 1

Step 2 of 6

Apply the Kirchhoff's current law at the node of voltage v_a .

$$\frac{v_a - 15.5 \times 10^{-3}}{10 \times 10^3} + \frac{v_a}{30 \times 10^3} + \frac{v_a - v_1}{20 \times 10^3} = 0$$

$$v_a \left(\frac{1}{10 \times 10^3} + \frac{1}{30 \times 10^3} + \frac{1}{20 \times 10^3} \right) + v_1 \left(\frac{-1}{20 \times 10^3} \right) = \frac{15.5 \times 10^{-3}}{10 \times 10^3}$$

$$v_a \left(\frac{11}{60 \times 10^3} \right) + v_1 \left(\frac{-1}{20 \times 10^3} \right) = \frac{0.0155}{10 \times 10^3}$$

$$v_a \left(\frac{11}{6} \right) + v_1 \left(\frac{-1}{2} \right) = 0.0155 \dots\dots (1)$$

Step 3 of 6

Determine the voltage at the non-inverting terminal.

Consider that no current flows into the terminals of an ideal op-amp.

Apply the voltage division principle.

$$v_1 = \left(\frac{80}{80+20} \right) v_b$$

$$= 0.8v_b$$

Recall equation (1).

$$v_a \left(\frac{11}{6} \right) + v_1 \left(\frac{-1}{2} \right) = 0.0155$$

Replace v_1 by $0.8v_b$.

$$v_a \left(\frac{11}{6} \right) + (0.8v_b) \left(\frac{-1}{2} \right) = 0.0155$$

$$v_a \left(\frac{11}{6} \right) + v_b (-0.4) = 0.0155$$

$$v_a = \frac{0.0155 + 0.4v_b}{\left(\frac{11}{6} \right)} \dots\dots (2)$$

Step 4 of 6

Apply the Kirchhoff's current law at the node of voltage v_b of the circuit shown in figure 1.

$$\frac{v_b - 15.5 \times 10^{-3}}{40 \times 10^3} + \frac{v_b}{60 \times 10^3} + \frac{v_b - v_1}{20 \times 10^3} = 0$$

$$v_b \left(\frac{1}{40 \times 10^3} + \frac{1}{60 \times 10^3} + \frac{1}{20 \times 10^3} \right) + v_1 \left(\frac{-1}{20 \times 10^3} \right) = \frac{15.5 \times 10^{-3}}{40 \times 10^3}$$

$$v_b \left(\frac{11}{120 \times 10^3} \right) + v_1 \left(\frac{-1}{20 \times 10^3} \right) = \frac{0.0155}{40 \times 10^3}$$

$$v_b \left(\frac{11}{6} \right) + v_1 (-1) = \frac{0.0155}{2}$$

Replace v_1 by $0.8v_b$.

$$v_b \left(\frac{11}{6} \right) + (0.8v_b)(-1) = \frac{0.0155}{2}$$

$$v_b \left(\frac{11}{6} \right) + v_b (-0.8) = \frac{0.0155}{2}$$

$$v_b \left(\frac{31}{30} \right) = \frac{0.0155}{2}$$

$$v_b = 7.5 \text{ mV}$$

Step 5 of 6

Recall equation (2).

$$v_a = \frac{0.0155 + 0.4v_b}{\left(\frac{11}{6}\right)}$$

Substitute 7.5×10^{-3} for v_b .

$$\begin{aligned} v_a &= \frac{0.0155 + 0.4(7.5 \times 10^{-3})}{\left(\frac{11}{6}\right)} \\ &= 10.09 \text{ mV} \end{aligned}$$

Step 6 of 6

Apply the Kirchhoff's current law at the node of voltage v_1 of the circuit shown in figure 1.

$$\frac{v_1 - v_a}{20 \times 10^3} + \frac{v_1 - v_o}{80 \times 10^3} = 0$$

Replace v_1 by $0.8v_b$.

$$\frac{0.8v_b - v_a}{2} + \frac{0.8v_b - v_o}{8} = 0$$

$$0.4v_b - \frac{1}{2}v_a + 0.1v_b - \frac{1}{8}v_o = 0$$

$$v_o = 8(0.5v_b - 0.5v_a)$$

Substitute 10.09×10^{-3} for v_a , and 7.5×10^{-3} for v_b .

$$\begin{aligned} v_o &= 8 \left[0.5(7.5 \times 10^{-3}) - 0.5(10.09 \times 10^{-3}) \right] \\ &= 8(3.75 \times 10^{-3} - 5.045 \times 10^{-3}) \\ &= -10.36 \text{ mV} \end{aligned}$$

Therefore, the value of voltage v_o is -10.36 mV .